

Project Initialization and Planning Phase

Date	12 July 2026
Team ID	XXXXXXX
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	The primary objective is to develop an AI-based system that predicts an individual's creditworthiness using Machine Learning, enabling users to understand their loan eligibility and receive personalized financial insights.
Scope	The project includes data collection, preprocessing, feature selection, machine learning model development, prediction, visualization of results, and deployment through a web application for real-time user interaction.
Problem Statement	
Description	Many applicants find it difficult to obtain quick loan decisions because existing systems are opaque, manual, and rely on rigid, outdated criteria.
Impact	Solving this problem will enable users to receive faster loan decisions, increase transparency in the application process, and allow lenders to make more accurate risk assessments.
Proposed Solution	
Approach	Develop a Machine Learning model (using algorithms like Random Forest or XGBoost) to analyze applicant financial and personal data, predict creditworthiness, and display the results through a web interface with graphical visualizations.
Key Features	• User-friendly web interface using Flask. • Secure user data collection and validation. • Data preprocessing and feature selection. • Well-being prediction using Linear Regression. • Interactive dashboard with graphs and charts. • Personalized recommendations for improving well-being. • Model saving and real-time prediction using Joblib.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU for model training and prediction	Intel Core i5/i7 or equivalent
Memory	RAM for processing datasets	Minimum 8 GB RAM
Storage	Space for datasets, trained model, and project files	256 GB SSD or higher
Software		
Frameworks	Backend Framework	Flask
Libraries	Machine Learning & Data Processing	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development Environment	IDE	Visual Studio Code / Jupyter Notebook
Data		
Data	Source, size, format	Well-Being Dataset (CSV format), containing health, lifestyle, and demographic features used for training and testing the Machine Learning model.